KIM, TAEHYEUN

Department of Aerospace Engineering, University of Michigan, Ann Arbor, MI

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Citizenship Status: U.S. Citizen

Research Interest

Safety-Critical Learning-based Control; Spacecraft Rendezvous, Proximity Operations, and Docking (RPOD)

Education

The University of Michigan Ann Arbor Expected 2027
Ph.D. in Aerospace Engineering, Advisors: Prof. Ilya Kolmanovsky and Prof. Anouck Girard

The University of Colorado Boulder May 2022
M.S. in Aerospace Engineering and Sciences

Yonsei University, Seoul, Korea Aug. 2020
B.S. in Astronomy ***(*****Class Rank: 1/35*****)***

Publications

Journal

[1] **T. Kim** and I. Kolmanovsky, "Filter-Aware Optimization of Cislunar PNT Constellations Across CR3BP Orbit Families," under review in the IEEE Transactions on Aerospace and Electronic Systems.

[2] **T. Kim**, T-G. Kim, A. Girard, and I. Kolmanovsky, "Learning Unknown Gravitational Fields for Spacecraft State Estimation: Temporal Hamiltonian Neural Network-based Adaptive Kalman Filtering," under review (revision submitted) in the IEEE Transactions on Aerospace and Electronic Systems.

[3] **T. Kim**, A. Girard, and I. Kolmanovsky, "Shrinking Horizon MPC for Safe Multi-Spacecraft Capture of a Non-Cooperative Target," IEEE Control Systems Letters, 2025, 2915-2920.

[4] **T. Kim**, A. Girard, and I. Kolmanovsky, "Constrained Spacecraft Rendezvous on Elliptic Orbits Using Time Shift Governors," Journal of Guidance, Control, and Dynamics, 2025, 1-11.

Conference Proceedings

[1] R. Kee***, T. Kim***, A. Girard, and I. Kolmanovsky, "Time Shift Governor-Guided MPC with Collision Cone CBFs for Safe Adaptive Cruise Control in Dynamic Environments," 2025 IEEE Conference on Control Technology and Applications.

[2] **T. Kim**, A. Girard, and I. Kolmanovsky, "CIKAN: Constraint Informed Kolmogorov-Arnold Networks for Autonomous Spacecraft Rendezvous using Time Shift Governor," Learning for Dynamics and Control 2025.

[3] **T. Kim**, A. Girard, and I. Kolmanovsky, "Autonomous Spacecraft Proximity Operations in the Earth-Moon Cislunar Space Using Time Shift Governors," AIAA SCITECH 2025 Forum, 0311.

[4] **T. Kim***, R. Kee*, I. Kolmanovsky, and A. Girard, "Constrained Control for Autonomous Spacecraft Rendezvous: Learning-Based Time Shift Governor," AIAA SCITECH 2025 Forum, 1700.

[5] S. Henao-Garcia, M. Kapteyn, K. Willcox, M. Tezzele, M. Castroviejo-Fernandez, **T. Kim**, M. Ambrosino, I. Kolmanovsky, H. Basu, P. Jirwankar, R. Sanfelice, "Digital-Twin-Enabled Multi-Spacecraft On-Orbit Operations," AIAA SCITECH 2025 Forum, 1432.

[6] **T. Kim**, I. Kolmanovsky, and A. Girard, "Time Shift Governor for Constrained Control of Spacecraft Orbit and Attitude Relative Motion in Bicircular Restricted Four-Body Problem," 2024 American Control Conference.

[7] **T. Kim**, I. Kolmanovsky, and A. Girard, "Time Shift Governor for Spacecraft Proximity Operation in Elliptic Orbits," AIAA SCITECH 2024 Forum, 2452.

[8] **T. Kim**, K. Liu, I. Kolmanovsky, and A. Girard, "Time Shift Governor for Constraint Satisfaction during Low-Thrust Spacecraft Rendezvous in Near Rectilinear Halo Orbits," 2023 IEEE Conference on Control Technology and Applications, 418-424.

[9] **T. Kim**, S. Boone, and J. McMahon, "Higher-Order Feedback Control Law for Low-Thrust Spacecraft Guidance," AAS/AIAA Astrodynamics Specialist Conference, 2022.

Research Experience

Graduate Student Research Assistant, University of Michigan, Ann Arbor, MI, USA Jan. 2022-Present
Vehicle Optimization, Dynamics, Control and Autonomy (PI: Prof. Ilya Kolmanovsky and Anouck Girard)

- Develop constrained control algorithms, including shrinking horizon MPC and time shift governor, for on-orbit spacecraft

- servicing under Air Force Office of Scientific Research (AFOSR) Space University Research Initiative (SURI).
- Formulate and solve an optimization problem for position, navigation, and timing systems in cislunar space under US Space Force & Air Force Research Laboratory (AFRL) xRADAR program.
- Develop control and navigation algorithms for constrained spacecraft rendezvous and docking, using numerical optimization and learning-based approaches.

Research Assistant, University of Colorado, Boulder, CO, USA Dec. 2021-Aug. 2022
Orbital Research Cluster for Celestial Applications (PI: Prof. Jay McMahon)
Developed spacecraft on-board guidance algorithms to improve their robustness for converging on solutions based on state transition tensors (STTs) and higher-order differential dynamic programming (DDP).

System Engineer, University of Colorado, Boulder, CO, USA Dec. 2021-Aug. 2022
Space Weather Atmospheric Reconfigurable Multiscale Experiment (PI: Prof. Scott Palo)
Project: Design and built three 3U CubeSats in LEO to investigate the relationship between charged particles and neutral particles.
1) Identified the correlation between the Dilution of Precision and the GNSS-Zenith angle during OGNC experiments in MATLAB.
2) Simulated and analyzed the spacecraft's relative motion using Differential Drag control in MATLAB and Julia.

Research Intern, Satrec Initiative, Daejeon, Republic of Korea July 2021
Developed image acquisition planning algorithms to optimize a large area coverage from multiple satellite sensors using Mixed Integer Programming.

Teaching Experience

Graduate Student Instructor, University of Michigan, Ann Arbor, MI, USA Winter 2025
AEROSP 548: Astrodynamics (Prof. Ilya Kolmanovsky)

Graduate Student Instructor, University of Michigan, Ann Arbor, MI, USA Fall 2024
AEROSP 470: Control of Aerospace Vehicles (Prof. Ilya Kolmanovsky)

Grader, University of Colorado, Boulder, CO, USA Winter 2022
ASEN 6020: Optimal Trajectory (Prof. Daniel Scheeres)

Awards & Honors

Honors Student Fall 2017
Awarded by Yonsei University, Korea

Certificate of Appreciation Apr. 2017
Awarded by Joseph C. Holland, Colonel, AR, Commanding, U.S. Army

Certificate of Commendation Apr. 2017
Awarded by Suk-Jae Lee, the 8th Army Republic of Korea Army Support Group Commander

Certificate of Commendation Mar. 2017
Awarded by Jong-Wook Kim, Chairman, KATUSA Veterans Association Inc.

Certificate of Achievement Mar. 2017
Awarded by Brady A. Gallagher, LTC, AV, Commanding, U.S. Army

Scholarship

The Scholarship for Science (Half-tuition) Spring 2019
Awarded by Korea Center of Creativity and Personality, Korea

The Scholarship for Science (Half-tuition) Spring 2019
Awarded by Korea Student Aid Foundation, Korea

The Scholarship for Science (Full-tuition) Fall 2017
Awarded by Hanyang Human Resource Development Institute, Korea

Other Experience

Volunteer Teacher, Ephphatha Welfare Center for the Disabled, Pyeongtaek, Korea Nov. 2016-Feb. 2017
Conducted weekly English tutoring to socioeconomically disadvantaged groups.
Evaluated students' understanding of course materials and developed strategies to improve their learning.

Sergeant, Korean Augmentation To the United States Army (KATUSA), Pyeongtaek, Korea July 2015-Apr. 2017
S1 (Personnel), 3-2 General Support Aviation Battalion, 2 CAB, 2ID stationed at USAG Humphreys

Student Volunteer, International Yonsei Community (IYC), Seoul, Korea Spring 2014-Aug. 2022
Collaborated with the members to coordinate Korean cultural programs supporting foreign students familiarizing themselves with Korean culture and language.

Part-Time Teacher, Oryu High school, Seoul, Korea Jan. 2013-Feb. 2013

Developed and implemented 'Self-Directed Learning Program' by which could improve their own learning.

Skills & Activities

Technical Skills	Julia, Python, PyTorch, MATLAB, General Mission Analysis Tool, Satellite Tool Kit
Activities	IEEE Membership (student)
	AIAA Membership (student)
Certificates	STK Certificate, Export Compliance
Languages	English (Fluent), Korean (Fluent)